Xiongyule Zhou

jensenzhou47@gmail.com

Beijing Jiaotong University, No. 3
Shangyuancun, Haidian District, Beijing 100044,
China

EDUCATION

Beijing Jiaotong University, Beijing, China Sep 2023 – Jul 2027 (Expected)

Bachelor of Engineering in Mechatronics Engineering

Cumulative GPA: **3.67/4.0** (88.2/100)

Relevant Coursework: Artificial Intelligence, Robotics and Flexible Automation, Control Theory, Embedded Systems, Signals and Systems, Intelligent Mechatronic System Design

RESEARCH EXPERIENCE

Learning-Based Robotic Manipulation, Beijing Jiaotong University Sep 2025 – Feb 2026

- Developed a Python/PyTorch learning pipeline for a simulated robotic arm, using MATLAB/Simulink to model robot dynamics for grasping, target placement and object-handoff tasks.
- Applied DINOv2 and OpenCV-based visual representations to extract object-level features for vision-guided grasping, placement and handoff, increasing full-task success rate from 31.4% to 77.1%.
- Developed reinforcement learning pipelines for robotic manipulation, exploring policy optimisation methods such as PPO and SAC for simulated grasping and object-handoff tasks, improving task execution speed by 46.2%.

Autonomous Robot Path Planning and Control, Beijing Jiaotong University Jun 2025 – Aug 2025

- Implemented sensor acquisition, finite-state-machine logic and PID motor control on an STM32, integrating and testing these functions with host-computer perception and navigation modules for autonomous route execution.
- Integrated colour, ultrasonic and other sensors, developing feedback-based decision and control logic for line following and obstacle avoidance.
- Tuned sensing thresholds and PID gains through repeated track tests, coordinating perception, planning and control for autonomous navigation and obstacle avoidance. Increased driving speed by 61.1% over the initial baseline and achieved a 93.1% obstacle-avoidance success rate.

Thermoelectric Autonomous Vehicle Design, Beijing Jiaotong University Nov 2024 – Apr 2025

- Designed a thermally powered autonomous vehicle using SolidWorks, developing cam and gear transmission mechanisms for thermal-to-mechanical energy conversion.
- Integrated line-following hardware and completed mechanical assembly, calibration and system-level debugging, improving vehicle stability and reliability and achieving an 87.3% task success rate.

Reconfigurable Land–Air Amphibious UAV, Beijing Jiaotong University Mar 2024 – Aug 2024

- Developed and debugged the UAV flight-control system, refining attitude stabilisation, takeoff and landing control, and PID parameters through repeated whole-platform testing.
- Implemented and debugged ground–flight mode-switching logic, coordinating DC hub motors, four independently actuated servos and brushless propulsion motors for safe and responsive transitions between terrestrial driving and aerial flight.

WORK EXPERIENCE

Software Engineer Intern —Embodied AI, Anyverse Technology Co., Ltd. Aug 2026 – Present

- Developed an automated execution, AI-based motion detection, evaluation and feedback-driven iteration workflow for the 100 household tasks in the 2026 BEHAVIOR Challenge, reducing review time by 60.1%; the team ultimately achieved world-leading results in the competition. In an archived comparison of 50 matched repeated instances across 20 tasks, mean goal-satisfaction score increased from 0.1145 to 0.2315.
- Refined C++ motion control and trajectory planning for an in-house 7-DoF robotic arm, implementing forward and inverse kinematics (FK/IK) and five standard paths: straight-line motion along the X, Y and Z axes, square paths and circular paths. Enabled storage and cyclic execution of up to 64 taught waypoints. In limited real-robot tests of these paths, the largest measured peak joint-tracking error was approximately 0.12°.
- Optimised mixed-precision inference and inference backends for robot policies, evaluating PyTorch, TensorRT and graph-optimisation configurations. In matched-input offline experiments with four camera inputs and a 26-dimensional action output, reduced median prepared-call latency from 255.51 ms to 175.27 ms—a 31.40% reduction—while jointly assessing action-output deviations, end-effector pose differences and GPU memory usage.
- Extended a humanoid robot's C++/Python control SDK, developing over 30 interfaces and writing drivers for encoder-state access, PVT trajectory execution, trajectory export, dynamics and controller diagnostics to support real-robot experiments.
- Designed and conducted experiments on continuous robot-camera sampling and image–joint-state pairing, completing 49 sampling tests totalling 673.10 camera-stream hours and analysing 24 sets of image and joint-state recordings. Offline evaluation of nearest-timestamp pairing across 78,825 valid image frames yielded median absolute time differences of 2.43–2.59 ms, providing a quantitative basis for constructing visual and proprioceptive observations.

Human Biomechanics & Embodied Simulation Intern, Chongqing Xingshijie Co., Ltd. Jan 2026 – Mar 2026

- Built human musculoskeletal models in OpenSim and imported human-motion datasets, analysing over 100 model data sources covering 18 types of human movement to establish a simulation workflow for whole-body motion and joint biomechanics.
- Computed and analysed joint forces, loads and kinematic profiles across 156 motion sequences and 23 major joints, identifying high-load phases and biomechanical patterns relevant to movement assessment and injury prevention.
- Translated simulation results into human-motion and biomechanical priors for embodied AI, providing joint-level references for human motion understanding, humanoid motion representation and human-like robot control.

Robot Learning and Manipulation Intern, Csoft Jul 2024 – Aug 2024

- Built a ROS/Ubuntu ALOHA bimanual robotic-arm system for teleoperation-based human-action imitation, supporting object grasping, transfer, book handling and water pouring, with a 77.3% task success rate across 220 real-robot trials.
- Integrated dual-arm control, teleoperation, joint-state feedback, calibration and demonstration execution into an end-to-end manipulation pipeline, completing 36 sets of calibration and demonstration-execution tests to reproduce human-operated behaviours.
- Diagnosed hardware–software synchronisation and calibration errors through repeated real-robot trials, improving execution consistency for contact-rich grasping, transfer and pouring tasks and reducing the number of task failures during repeated execution by 28.6%.

SELECTED RESEARCH MANUSCRIPTS

- **Field-Specific Evidence Contracts for Dual-Arm Robot Feedback under Explicit Error Budgets.**
  Xiongyule Zhou. Journal of Intelligent & Robotic Systems. Under review.
  Developed a field-specific feedback validation method with explicit error budgets and evaluated it using dual-arm robot measurements.
- **Latency, Numerical Fidelity, and Resource Tradeoffs in Mixed-Precision Robot Policy Inference.**
  Xiongyule Zhou. IEEE Access. Under review.
  Compared mixed-precision and inference backends for robot policies, analysing latency, action fidelity and nominal end-effector deviation.
- **Pose coverage and held-out image validation for multi-camera robot calibration.**
  Xiongyule Zhou. Industrial Robot. Under review.
  Evaluated pose-selection strategies and calibration solvers using held-out images from multi-camera robot experiments.

AWARDS

- Second Prize, Beijing College Students Engineering Practice and Innovation Competition Mar 2025
- Third-Class Academic Excellence Scholarship 2024-2025, 2025-2026
- Third-Class Scholarship for Outstanding Social Work 2024-2025, 2025-2026
- Second-Class Sports Activity Scholarship 2023–2024

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Captain, Basketball Team, School of Mechanical Engineering Jun 2024 – Present

- Managed team operations, scheduling, and communication among players and coaching staff.
- Led the team to a top-four finish in the university basketball tournament.

Deputy Director, Visual Arts Department, University Arts Center Oct 2024 - Oct 2025

- Coordinated campus visual-media projects and produced photography/video content for university events.

SKILLS AND INTERESTS

Programming Languages: Python, C++, C, MATLAB
Robotics: ROS/ROS 2, Robot Manipulation, Motion Planning, Motion Control, Sim-to-Real
AI / ML: PyTorch, Deep Learning, Robot Learning, Computer Vision
Tools: Linux, Git, MATLAB/Simulink, MuJoCo / Isaac Sim
Languages: Chinese (Native), English (IELTS: 7.5)
Interests: Basketball (University Team Captain), Photography, Video Production, Reading, Robotics, Artificial Intelligence